

A Mathematical Theory of Retention

Records under construction: what a dynamics keeps, what it costs to write, and what it costs to keep

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Introduction

The problem of how a physical system retains information has been attacked from several directions, and this paper stands on all of them. Shannon (1948) gave the chain rule that splits an entropy into the entropy of a partition and the entropy within it, and everything below is a reading of that identity. Crutchfield and Young (1989) took the decisive step of letting the *dynamics* choose the partition rather than the modeller, defining the causal states and identifying the statistical complexity with their entropy; the ergodic decomposition of such quantities is treated by Crutchfield and Debowski (2006). Gell-Mann and Lloyd (1996) posed the same split as effective complexity. Cheeger’s inequality supplies the relation between a bottleneck and a spectral gap that we use to price a record. Tononi’s integrated information theory prices a whole against its best partition — the “cruellest cut” — which is the same construction we reach in Section 6 by another road. Eigen’s error threshold is the phenomenon our Section 7 recovers as a melting law. Bennett’s logical depth and, more recently, the assembly index of Cronin and Walker measure construction history, which we show in Section 9 is orthogonal to the quantity treated here rather than a competitor to it.

We say this at the front because the honest fraction of what follows that is new is small. Our estimate is about one part in five. What we believe is ours is this: the construction axis – what happens to a record’s class structure when the state space itself is *built* rather than held fixed (Section 5); the generator law that a record is destroyed by which moves a set contains and not how many (Section 7), together with the abelian/non-abelian split that explains why some systems melt linearly and others fall off a cliff; the coupling/selection ledger of Section 8; and the measured orthogonality of record and depth in Section 9. The rest is a re-derivation of known material in one vocabulary. We think the single ledger earns its keep. We do not think it is new physics.

Part I. The record system

1. Definitions

Definition 1. A *record system* is a pair (S, M) where S is a finite set of states and M is a set of involutions on S . The *move graph* is the undirected graph on S with an edge $\{s, ms\}$ for each m in M and s in S with $ms \neq s$. A *class* is a connected component of the move graph.

Classes are the ergodic components of the dynamics generated by M . A state factorises as (class, position within class): the class is what the dynamics cannot change and the position is what it can.

Definition 2. For the uniform measure on S , write

R = the Shannon entropy of the class distribution (the *record*), H_{thermo} = the mass-weighted mean of $\log_2(\text{class size})$ (the *heat*), $H = \log_2 |S|$.

Proposition 1. $H = R + H_{\text{thermo}}$.

Proof. The chain rule $H(X) = H(f(X)) + H(X | f(X))$ with f the class map. QED

Proposition 1 is Shannon's, restated. It is the accounting identity on which everything rests, and the reader should keep in mind that several of the theorems below are one line from it.

Definition 3. A *construction* is a map A from a record system (S, M) to a record system (S', M') . A construction *merges* if two parent classes map into one child class, and *splits* if one parent class maps onto more than one. A construction *retains* if it does not merge.

2. A warning about R

R is maximised when nothing moves. If every class is a singleton then $R = \log_2 |S|$ and $H_{\text{thermo}} = 0$, and the system is a perfect record that cannot do anything. R must never be read alone. The honest state of a record system is the pair (R, H_{thermo}) , and a trajectory that raises R while driving H_{thermo} to zero is not accumulating a record; it is crystallising. We return to this in Section 8 and it is the single most important caveat in the paper.

Part II. Retention

3. The retraction theorem

Theorem 1. Let $A : S \rightarrow S'$ be a construction and $r : S' \rightarrow S$ satisfy $r \circ A = \text{id}$. If every move of M' , seen through r , is a composition of moves of M , then A retains.

Proof. Suppose s, t lie in different parent classes but $A(s), A(t)$ in one child class. Then a sequence of M' -moves carries $A(s)$ to $A(t)$. Applying r and using the hypothesis, the image of that sequence is a composition of M -moves carrying s to t , so s and t are in one parent class, contradiction. QED

The clause in bold is load-bearing and is the most frequently dropped hypothesis in this subject, including by us. A retraction alone is not enough. Every failure of retention we have measured is a failure of that clause and not of the retraction.

Theorem 2 (certificate). A retains if and only if the child carries a function, constant on child classes, whose pullback along A separates the parent classes.

Proof. If A retains, the parent class map factors through the child class map and the child class label is such a function. Conversely such a function separates the images of distinct parent classes, so no two can lie in one child class. QED

Theorem 3. If the parent move graph is an induced subgraph of the child move graph on the image of A , then A retains, under any extension rule.

Proof. Immediate from Theorem 1 with r the restriction. QED

Theorem 3 is why crystals work. A growing lattice reproduces its own move structure inside the larger one, so a dislocation, a stacking fault or a twin boundary propagates indefinitely and cannot be destroyed by further growth. Retention, in this setting, is free.

Part III. Construction and creation

4. Accretion

Theorem 4. Let A adjoin a new site and let A be *total*, meaning every parent state extends. Then A induces a bijection on classes: it neither merges nor splits. Consequently a class is created if and only if the exclusion of the non-extending states disconnects a parent class.

Proof. Totality makes the class map surjective; Theorem 3 makes it injective; a bijection creates nothing. For the second claim, the child classes are the components of the parent move graph restricted to the extending states, so a parent class contributes more than one child class exactly when that restriction disconnects it. QED

Theorem 4 is the reason a crystal is not alive. Its growth is very nearly total, so it copies its record perfectly and forever and never writes a new one. Creation requires refusal.

5. The price of a bit

Theorem 5. Creating a new class from a parent class requires excluding a vertex cut of that class's move graph. Hence a bit costs at least the size of a minimum cut.

Proof. By Theorem 4 creation is disconnection, and to disconnect a connected graph one must remove a cut. QED

Corollary (why two). A move graph with no cut vertex admits no cut of size one. In every instance we have measured the move graph is 2-connected, so the minimum price of a bit is two.

This corollary has an unexpectedly exact echo. In the dimer instance a single forbidden bond never splits a class in 80 of 80 attempts – it only empties classes – while two-bond prohibitions split routinely (251 of 300 on one torus, 90.7 per cent of 300 on another). In the same instance a single cross bond between two copies is used by zero states, for a parity reason, while two are used freely. *One contact can neither cut nor couple; two can do both, and it is the same two.*

Remark 6 (formerly stated as a theorem). One expects the cut cost of Theorem 5 to be controlled by the spectral gap of the class's move graph, so that slow-mixing records are cheap to write and fast-mixing ones expensive. Cheeger's inequality bounds *edge* conductance, and Theorem 5 needs a *vertex* cut; the transfer requires a vertex-expansion inequality (Bobkov, Houdre and Tetali 2000) or a maximum-degree factor, and we have not carried it out. We keep the statement as a heuristic and number it so that later references resolve.

Part IV. The ledger

6. Selection and coupling

Theorem 7 (exchange). A purely selective step – one that removes states and adds none – has $\Delta H \leq 0$, hence $\Delta R \leq -\Delta H_{\text{thermo}}$. Since $R \leq H$, iterated pure selection terminates: R rises, reaches a maximum at or before the exhaustion of H_{thermo} , and thereafter can only fall.

Proof. From Proposition 1, $\Delta R = \Delta H - \Delta H_{\text{thermo}} \leq -\Delta H_{\text{thermo}}$. The terminal state of iterated selection has every class a singleton, $H_{\text{thermo}} = 0$, and every further step removes states and lowers R . QED

Theorem 7 is one line from Proposition 1 and we do not claim more for it. Its content is not the inequality but the regime: pure selection is *capacity-spending* and it ends as a crystal – maximum nominal record, nothing moving, and thereafter decline.

The measured rate is a property of the searcher, not of the space. We report this as a correction to our own earlier claim. On one torus the conversion ratio $\Delta R / (-\Delta H_{\text{thermo}})$ has median 0.517 under a greedy refusal policy, 0.556 under

exhaustive search, and 0.401 under a random policy; on the same states under a colder move set the three medians are 0.565, 0.565 and 0.223. Any constant read off a trajectory in this framework is a constant of the policy.

Coupling is the complementary operation. A step that adds contacts between two record systems is *capacity-creating*: it spends heat and opens space at the same time. Measured across two instances the conversion ratio is 1.29 to 2.11, always above one, with between 47 and 77 per cent of the record written taken out of H_thermo and the remainder out of newly opened space.

The ledger. Selection converts at most one and terminates. Coupling converts above one and does not. A system that alternates them accumulates; a system doing either alone does not.

We note explicitly what this does not say. Duplication is not free: two decoupled copies of a record system carry exactly twice its record, $R/2R(P) = 1.000$ to four decimal places in our measurement. Copying is cheap only once the copies are coupled.

Part V. What destroys a record

7. The generator law

The central experimental finding of this work is that a record is not destroyed by the *number* of available moves but by *which* moves are available. On one torus, ordering moves by cycle length gives a set of 154 moves under which $R = 0$; ordering the same moves by geometric diameter gives a set of 546 moves under which $R = 0.69$ bits survive. The larger set retains and the smaller one does not.

This is not explained by saying that the fatal moves “move the certificate”. Classes are *defined* as components of the move graph and a certificate is *defined* as a function separating them, so that statement unpacks to “a move that connects two classes merges them”, which is definitional. We flag it because we made exactly that error in an earlier draft. The non-vacuous content is quantitative, and requires knowing the shifts.

Theorem 8 (codimension). Let S carry a free transitive action of a finite abelian group V , with every move acting as translation by some v_m in V , and let μ be uniform. Then the classes are the cosets of $\Lambda = \langle v_m : m \text{ in } M \rangle$ and

$$R = \log_2 |V / \Lambda|,$$

exactly. For $V = F_2^n$ this is $n - \text{rank}(\Lambda)$.

Proof. The orbit of s under the group generated by the moves is $s + \Lambda$, a coset, and a free transitive action makes every coset the same size, so the uniform measure on S pushes to the uniform measure on V / Λ . QED

Remark (the torsion-free case is a codimension, not a bit count). Theorem 8 is stated for finite V . For $V = \mathbb{Z}^n$ – a lattice of chemical compositions moved by the columns of a stoichiometric matrix S – the compositions are not a torsor and V / Λ is infinite whenever $\text{rank}(\Lambda) < n$, so $\log_2 |V / \Lambda|$ is infinite. The honest quantity there is the codimension $n - \text{rank}(S)$, which is the dimension coefficient of Section 9a and equals the number of independent conservation laws (the conserved moieties of reaction-network theory: Schuster and Hofer 1991; Feinberg 2019). The companion paper’s “one reaction is one bit” is one dimension of codimension, and its integer checks are rank-nullity verified numerically – algebra, not measurement.

Theorem 8 is the abelian case. The next theorem is what happens when the action is not free, so that a fibre of $S \rightarrow V / \Lambda$ can hold more than one class.

Theorem 9 (certificate translation). Let $\phi : S \rightarrow V$ be a map to an abelian group V such that every move m in M acts on it by a translation, $\phi(ms) = \phi(s) + v_m$. Let $\Lambda = \langle v_m : m \in M \rangle$ be the subgroup generated by the shifts. Then every class lies inside a single fibre of the composite $S \rightarrow V \rightarrow V/\Lambda$, and

$$R = H(\phi_* \mu \text{ on } V / \Lambda) + H(\text{class} \mid \text{fibre}),$$

exactly. The second term is the *defect*. It vanishes if and only if every fibre is a single class. In particular a move with $v_m = 0$ enters no Λ and so cannot lower the fibre term, however many such moves are added; it may still merge classes within a fibre and so reduce the defect.

Proof. Along any move, ϕ changes by an element of Λ , so $S \rightarrow V/\Lambda$ is constant on move-orbits, i.e. on classes. The class partition therefore refines the fibre partition, and the chain rule for the entropy of a refinement gives $R = H(\text{fibre}) + H(\text{class} \mid \text{fibre})$. The defect vanishes exactly when no fibre splits, which is transitivity of the action on fibres. A move with zero shift lies in the kernel of every such quotient and so enters no Λ ; nothing prevents it from joining two classes that share a fibre. QED

An earlier draft printed the last clause as “cannot merge any classes: it is free.” That is false, and the paper’s own data is the counterexample: the 24 face flips of the 4x6 dimer torus all have shift $(0, 0)$, and they take the system from $R = \log_2 |S|$ with no moves to $R = 2.2572$. What a zero-shift move cannot do is lower the certificate bound; “free” below means free relative to that bound.

The defect is the record the certificate cannot see, and it is measurable. Verified exactly (machine precision) on five systems: 4x6 faces $2.2572 = 2.2572 + 0.0000$; 4x6 all-4-cycles $0.8240 = 0.6908 + 0.1332$; 4x4 faces $2.3159 = 2.2865 + 0.0294$; 4x4 all-4-cycles $0.2797 = 0.0000 + 0.2797$ (the certificate is worthless there and every bit of record is hidden from it); 6x6 faces $2.3504 = 2.3500 + 0.0004$. The tightness condition is a pure connectivity check on the move graph – no entropies need be computed to apply it.

Corollary (why the abelian case is exact). If the moves generate an abelian group

acting by translations, the cosets ARE the orbits, so the defect is identically zero and R is exactly the codimension of Section 7's Theorem 8. This is why the meiotic instance of the companion paper satisfies $R = (L-1) - |J|$ at every value tested, while no dimer instance can.

Theorem 9 is what makes Section 7's experiment non-trivial, because the bound is computable from the shifts alone, without ever constructing the move graph. Two checks:

- Meiosis (equality holds). $V = F_2^n$, a crossover at junction j has shift e_j , so Λ is the span of the accessible junctions and V/Λ has $2^{(n - |J|)}$ equally-weighted elements. The theorem gives $R = n - |J|$ exactly, matching the measured value at all seven values tested.
- Dimers (nonzero defect). $V = \mathbb{Z}^2$, the winding number. Every one of the 24 face flips has shift $(0,0)$, so they leave the fibre term at its no-move value, and the fibre entropy computed from the shifts alone reproduces the measured $R = 2.2572$ to four decimal places with no reference to the move graph. The 15 wrapping cycles have shifts $(0, +4)$, generating $\Lambda = 4\mathbb{Z}$; the fibre term falls to 0.6908 and the defect accounts for the remaining 0.1332 exactly. The subgroup is identified correctly and the decomposition closes.

The empirical statement of Section 7 is therefore explained without tautology: the 546 compact moves all have zero shift and are free relative to the certificate bound – they merge classes within fibres, which is why 0.69 bits and not 2.26 survive – while a single wrapping cycle enlarges Λ and collapses the bound itself. What matters is the subgroup the shifts generate, not the cardinality of the move set.

8. What a junction is for

A junction between two record systems pays only if it blocks moves. We had claimed that complexity lives at the interface, and we withdraw the claim in that form. At matched substrate a coupled pair does carry more record than a monolith – a factor of 2.4 at matched state-space size, not the factor of 4 we first reported at matched vertex count – but the mechanism is that contacts occupy sites and thereby *forbid* moves that would otherwise mix. Where a junction adds material without adding constraint it adds nothing: at matched genetic material, outcrossing shows no advantage over selfing (ratio 0.77 to 1.07 across the recombination range).

On the number of parties. Holding substrate, unit count and contact budget all fixed, and varying only how the contacts are distributed, record rises monotonically as contacts are concentrated: three junctions 5.58 bits, two junctions 6.73, one junction 8.10. Three bodies forced into a single junction converge to the efficiency of two bodies (1.486 against 1.51). The cost is seams, not bodies. Two is then the smallest party that can cut at all, by the corollary to Theorem 5, and every party beyond two dilutes a fixed contact budget across more seams.

Part VI. The observer, and the temperature

8a. Refusal requires an observer

The three operations of Part IV are not symmetric, and treating them as though they were is an error we made and had to correct.

Growth and coupling are observer-free. Bonds form; vertices attach; nothing has to decide. Retention is observer-free. Theorem 1 is a property of maps. Refusal is not. To exclude some states and not others, something must distinguish them. That is a measurement, and a measurement requires an apparatus. The apparatus need not be conscious – a lion is an apparatus, so is a solvent, so is a hole of a given size – but it is not nothing.

The refusal operator is Maxwell’s demon, and Theorem 7 is Szilard’s accounting with the demon’s own cost omitted.

This is not a defect to be repaired but a fact to be recorded, and it explains an observation we first mis-filed as contamination. On one torus the conversion ratio has median 0.517 under a greedy refusal policy, 0.556 under exhaustive search, and 0.401 under a random policy; on the same states under a colder move set, 0.565, 0.565 and 0.223. We had reported 0.50-0.65 as a property of the record space. It is a property of the searcher. The demon’s efficiency depends on the demon.

Charging the demon changes the ranking. Taking $H_{\text{obs}} = \log_2(\text{candidates examined})$ as the memory that must be erased between cycles, the net conversion becomes 0.129 for greedy, 0.103 for exhaustive and 0.401 for random: the cheap searcher wins. We flag that this accounting is not settled – a comparison-based argmax can in principle run reversibly, holding only a running best – but the term exists and is the same order as the heat term.

What a receptor costs. The cheapest demon is one that does not measure and then decide, but in which the measurement IS the decision: a binding site, a hole of a given size, a flow that sorts by speed. Nothing is stored, so nothing is erased, and the per-read cost is zero. The price is instead the standing maintenance of the distinguisher’s shape – which is $R(T)$ for the distinguisher itself, so the observer enters this ledger as another record rather than as an external term. Optimising selectivity b against a mistake cost c and a record decay rate μ gives an interior optimum $b^* = \log_2(c \ln 2 / \mu)$, independent of the value of good outcomes: a system discriminates because mistakes are expensive, not because the prize is good. Above $\mu = c \ln 2$ no maintained specificity pays at all and the optimal receptor is none, so geometric selection – which holds no record, because it IS the shape – necessarily precedes chemical selection.

8b. $R(T)$: the record an observer of finite patience can see

A record system whose own state gates its moves forces a change in the definition of R . If the gating can be rewritten freely, every state is reachable from every other,

the whole space is one class, and $R = 0$: a record that can freely unshelter itself has none. Shelter therefore requires a rate hierarchy – the gate must be slower than what it gates – and R must be indexed by how long one is willing to watch.

Definition. $R(T)$ is the record computed using only those moves whose rate exceeds $1/T$. It is the record visible to an observer who watches for time T .

Two consequences. First, this is the canonical ordering of move sets. Section 10 reports that no cost function on moves gives a convention-free melting threshold. Rates are not a cost function chosen by the modeller; they are physical, and ordering moves by rate is therefore canonical. The observer question and the temperature question are the same question, and $R(T)$ answers both. Second, $R(T)$ is a step function – flat plateaus and sudden one-bit drops as each generator’s rate crosses the threshold, with no exponential attrition anywhere. Smooth erosion and sudden loss are different physics: attrition is ongoing mixing, a sudden loss is a generator activating.

A system cannot be its own demon; two systems can be each other’s. Measured: two identical systems at equal rates hold 13.87 bit-decades under mutual gating and exactly 0.00 under one-way gating or none. Mutual gating manufactures a rate hierarchy between equals, and the advantage is largest precisely where the partners are most alike, vanishing when one is already 1000 times slower than the other. The mechanism is deadlock: 64 mutually-locked configurations, each partner holding the other still, embedded in a mobile bulk class of 192, with H_{thermo} staying at 5.69 of 8 so the assembly as a whole is not frozen.

Part VII. Limits

9. Record is not depth

Assembly-theoretic and logical-depth measures ask how many construction steps an object required. That is a different quantity from R and we can exhibit the independence. Two of our instances have the same record to within 0.035 bits – 2.3159 and 2.3504 – and mean class diameters of 7.65 and 25.45, a factor of 3.33. Within a single family the two rise *together* as the move set cools, contrary to our own prediction. And the frozen endpoint has R maximal at depth zero.

This theory is therefore not a reparameterisation of assembly theory, and neither measure subsumes the other. The honest complexity coordinate is at least the triple $(R, H_{\text{thermo}}, \text{depth})$.

9a. The continuum: the record is a dimension, not a count

We attempted a non-lattice instance – the Chirikov standard map – and failed twice, concluding that R might not be measurable off a lattice. That conclusion was wrong, and the reason is instructive. Both instruments counted connected components of

a coarse-grained phase space. But the invariant structure of a near-integrable map is a positive-measure Cantor set of tori, not a partition into countably many classes: counting components is simply the wrong functional, and no refinement of the grid repairs it.

In the continuum the orbit space is a manifold, so a count of classes diverges. Regularising at a resolution eps gives the Kolmogorov epsilon-entropy,

$$R(\text{eps}) \sim \dim(\text{orbit space}) * \log_2(1/\text{eps}) + O(1),$$

whose leading coefficient is the DIMENSION. The continuum analogue of the record is a dimension (or a measure), and $R(T)$ of Section 8b is precisely the regularisation that renders it finite. We measure it on the standard map as follows. Definition. Take initial conditions on a uniform 256×256 grid of cell centres in $[0, 2\pi]^2$, iterate $x' = x + y'$, $y' = y + K \sin x$ with y unwrapped, and call an initial condition *non-mixing at horizon T* if its orbit never moves a full period in y within T steps, $\max_{t \leq T} |Y_t - Y_0| < 2\pi$. The measure $m(K, T)$ is the fraction of initial conditions that are non-mixing. This is an $R(T)$ -type quantity and the horizon is part of the definition. Predictions were hashed before the run (toy 5691, digest cc492359).

K	T = 300	T = 1000	T = 3000	T = 10000
0 through $K_c = 0.9716$	1.0000	1.0000	1.0000	1.0000
1.0	1.0000	1.0000	1.0000	1.0000
1.2	0.9996	0.9810	0.8790	0.6297
1.5	0.7861	0.4852	0.3482	0.3169
2.0	0.3095	0.2661	0.2572	0.2466
3.0	0.1212	0.1198	0.1186	0.1176

Below K_c the measure is exactly one at every horizon, as a rotational KAM torus requires, and the $K = 0$ control is exact. Above K_c the measure decreases monotonically in K and in T , approaching at large K to the island measure and, just above K_c , depending strongly on the horizon because transport through the broken tori is slow. The decay is graded, not a one-generator cliff: every intermediate value is populated, which is the third category the abelian/non-abelian split of Section 7 needed. We had pre-registered a stronger reading, that no single step on the grid $K = 1.0, 1.2, 1.5, 2.0, 3.0$ carries more than half the total drop, and it held only at $T = 10000$; at the three shorter horizons the step from 1.2 to 1.5 carries more than half. “Smooth” is therefore a statement about the long-horizon limit, and at finite horizon the melting concentrates just above K_c . An earlier draft quoted four values here from a diagnostic run whose script was not retained and whose definition was not stated; those values do not agree with the present instrument at any horizon (maximum discrepancy 0.13 or worse) and are withdrawn.

(An earlier draft carried here a remark on the shared formalism of orbit quotients with the theory of homogeneous domains. It transferred no result and named an

object foreign to this paper, so it is removed; the observation lives in the working record.)

10. There is no canonical *cost*, but there is a canonical *clock*

Every quantity in this paper is stated relative to a move set. We first hoped the *melting threshold* would be intrinsic even if R was not, and it is not: two cost functions on the identical system place melting at incomparable move sets, and the larger set survives – 546 moves retain 0.69 bits where 154 retain nothing. No cost function chosen by the modeller yields a convention-free temperature.

The resolution is Section 8b. Cost functions are inventions; rates are physics. Ordering the move set by rate and indexing by the observer's horizon gives $R(T)$, which is canonical in the only sense available: it is relative to an observer, and the observer is part of the system rather than a convention imposed on it. This remains a thermodynamics of records relative to a stated move set. What Section 8b adds is that the relativity is to a *CLOCK*, not to a taste. The corollary is methodological and we think it is the most useful sentence in the paper: *the move set is not a nuisance parameter to be eliminated. It is the physics*. An application of this theory to any real system is not complete until the states, the moves and the construction step are named for that system, and we have done that here for exactly one – meiosis, in the companion paper.

11. A definition of living, offered as a target

Our first attempt – sustains $\Delta R > 0$, holds H_{thermo} above a floor, carries two certificates – is insufficient, because every one of those can be done TO a system from outside, and we did exactly that to a torus fourteen times in the course of this work without producing anything alive. The distinguishing property is not the trajectory but who performs the refusal.

A crystal makes no refusals. A selected population's refusals are made by its environment. A living system makes its own.

With that added, a record system is living, at a given move set, if it sustains $\Delta R > 0$ per construction cycle while holding H_{thermo} above a floor, carrying two or more certificates on disjoint supports, with the refusals performed from inside. "Inside" is not defined for a bare pair (S, M) ; this is offered as a target for a definition, not as one.

Where such systems sit. Two criteria give two different answers, and the difference is instructive. Under a STATIC definition of the living boundary – as near frozen as possible while still moving – the meiotic instance holds $H_{\text{thermo}} = 1.0000$ bit at every size from $n = 6$ to $n = 16$, one active generator and no more. But that exactness is an artefact of the static definition, not of the substrate: running a DYNAMIC criterion instead, and selecting trajectories by whether they SUSTAIN (R rising with H_{thermo} off the floor), the ratio becomes the tight quantity on both substrates while the absolute floor spreads.

substrate	H_thermo rel. sd	R/H rel. sd	sustaining R/H
dimers (coupled blocks)	0.499	0.108	0.791 +- 0.085
abelian (independent generators)	0.714	0.114	0.829 +- 0.095

Sustaining systems sit HIGH – above $R/H = 0.6$ in every case tested, never near the $1/2$ we predicted and never at either death. This is a ONE-SIDED BOUND, and we state it as no more than that. On the abelian substrate it is not even that: by the closed form below, $R/H > 0.6$ is $f < 0.4 + 1/g$, a property of the parameter grid we swept. Only the dimer row is evidence independent of the grid.

We withdraw a stronger claim we briefly made. Having found the sustaining values clustered at 0.79 and 0.83 with small standard deviations, we reported $R/H \sim 0.8$ as a candidate universal. It is not. On the abelian substrate the steady state has a closed form: with growth adding g junctions of which a fraction f are active, and refusal removing one active generator per step, $J/n \rightarrow f - 1/g$ and therefore $R/H \rightarrow 1 - f + 1/g$. That formula reproduces every measured value to a mean error of 0.023 and spans 0.667 to 1.000 across the parameter grid we swept, against a measured range of 0.661 to 0.956. The apparent clustering is the narrowness of our sweep. It is the same error that disqualified our uniform-sampling test, committed a second time in the experiment built to repair it, and we record it here rather than in a footnote because the failure mode is evidently easy to repeat: *a tight distribution over a swept parameter is a property of the sweep until shown otherwise*. What survives is the one-sided bound above, and the observation that two different substrates gave overlapping ranges – which is suggestive of substrate-independence and not, on this evidence, a demonstration of it. The definition earns its keep by the fact that each way of failing it names something we measured: *spent* (selection without growth: R peaks, H_{thermo} exhausts, R decays); *crystal* (growth without refusal: Theorem 4, $\Delta R = 0$ forever); *crystal at scale* (growth below the sustaining threshold: R reaches 99 per cent of its ceiling with H_{thermo} at 0.089); *melted* (a generator crosses the junction; the certificate fails at the first such move, not gradually); and *gas* (the move set is ergodic and $R = 0$). We offer it as a target to be attacked, not as a result.

Appendix A. Instances

Three record spaces were used. Dimer coverings of tori (4×4 , 4×6 , 6×6 ; up to 90,176 states) under alternating-cycle flips, with the winding number as certificate – exact on the bulk, with 16 classes at extremal winding that are frozen singletons holding 0.018 per cent of the mass. Proper colourings under Kempe moves. Meiotic phase under crossovers, treated in the companion paper. All results are exact enumer-

ations. Predictions were written and hashed before each run and the digests are retained.

Appendix B. Errors found and corrected in this work

We list these because the pattern is informative: in two days of work, the numbered theorems stood unchanged except one clause of Theorem 9, found on cold read by a colleague on the third day (a zero-shift move was said to be unable to merge classes; it cannot lower the fibre term, which is weaker), and Theorem 6 was demoted to a remark for applying an edge inequality to a vertex cut; six informal statements built on the theorems did not survive.

Withdrawn: that duplication is free in R; that the conversion rate 0.50-0.65 is a property of the record space (it is the greedy policy's); that H_{thermo} reaches zero exactly at the peak of R (true in one instance of two, corrected to “at or before”); that complexity lives at the interface without qualification (only where the junction blocks moves, and at $2.4x$ not $4x$); that a third party is “too complex” (it is under-used; the cost is seams); and that two competing curves do not cross (undetermined; read from one point).

Two instrument faults were caught by positive controls and are worth naming. A registration step that silently failed to execute because of a shell short-circuit, which is why every prediction file in this work is now hash-verified *before* the run rather than after. And a modelling error in which states were canonicalised by sorting, so that “the pair (0,1)” silently meant “the two lexicographically smallest” rather than a fixed pairing; the control that caught it reported zero of 41 states moved by a move that should have moved most of them.

A third, found on audit the next day: the standard-map figures of Section 9a were quoted from a diagnostic run whose script was never retained, with the quantity undefined. They were replaced by a registered instrument with the definition frozen before the run (Section 9a states both the definition and the horizon). A number without a retained instrument is a memory, not a measurement.

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Internal draft. Every entry is from the authors' memory and must be pinned to the primary source before any external copy.

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